The documentation and process conversion measures necessary to comply with this revision shall be completed by 15 Jan 93.

INCH-POUND

MIL-S-19500/124G
15 October 1992
SUPERSEDING
MIL-S-19500/124F
1 August 1980

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE REGULATOR B AND RB
TYPES 1N2970 THROUGH 1N2977, 1N2979, 1N2980, 1N2982, 1N2984
THROUGH 1N2986, 1N2988 THROUGH 1N2993, 1N2995, 1N2997, 1N2999
THROUGH 1N3005, 1N3007, 1N3008, 1N3009, 1N3011, 1N3012, 1N3014, 1N3015,
A AND RA TYPES 1N3993 THROUGH 1N3998, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 10 watt, silicon, voltage regulator diodes: A or B type (standard polarity); RA or RB type (reverse polarity). Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (DO-4).

1.3 Maximum ratings. (maximum ratings are as shown in columns 4, 8, and 10 of table V herein and as follows:

$-65°C \leq T_J \leq +175°C$; P_T = 10 W at T_C = +55°C; derate at .083 W/°C above +55°C.
$-65°C \leq T_{STG} \leq +200°C$.

1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in columns 2, 9, 12, and 14 of table V herein, and as follows:

Thermal resistance ($R_{\theta JC}$) = 12°C/W maximum.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command Electronics Technology and Devices Laboratory ATTN: SLCET-RS Fort Monmouth, NJ 07703-5601 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

AMSC N/A FSC 5961

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SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

a. Device types 1N2970B through 1N3015B and 1N3993RA through 1N3998RA shall have the anode connected to the stud.

b. Device types 1N2970RB through 1N3015RB and 1N3993A through 1N3998A shall have the cathode connected to the stud.

3.3.1 Lead finish. Lead finish shall be gold, silver, or tin-lead plated. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the diodes, but shall be retained on the initial container.

3.4.1 Reverse polarity types. Reverse polarity units (see 3.3a and b) shall be marked with an "R" preceding the "A" or "B" in the type designation, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
7(b)	MIL-STD-750, test method 1071, condition C, step 2.	MIL-STD-750, test method 1071, condition C, step 2.
9	I_R and V_Z (for devices with $V_{Z(nom)} \geq 10$ V dc; see column 2 of table V).	Not applicable
11	I_R and V_Z; ΔI_R = 100% of initial value or 1 of column 12 of table V, whichever is greater; ΔV_Z = ±2.5% of initial value (for devices with $V_{Z(nom)} \geq 10$ V dc; see column 2 of table V).	I_R and V_Z
12	See 4.3.1	See 4.3.1
13	Subgroups 2, 3, and 4 of table I herein; ΔI_R = 100% of initial value or 1 of column 12 of table V, whichever is greater; ΔV_Z = ±2.5% of initial value.	Subgroup 2 of table I herein; ΔI_R = 100% of initial value or 1 of column 12 of table V, whichever is greater; ΔV_Z = ±2.5% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

I_Z = Column 15 of table V; V_Z = Column 2 of table V; T_C = 150 ±5°C.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. End point electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV), of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 2 A dc	V_F		1.5	V dc
Reverse current	4016	V_R = column 11 of table V, dc method	I_{R1}		Column 12	A dc
Regulator voltage (see 4.5.3)	4022	I_Z = column 5 of table V	V_Z	Column 3	Column 4	V dc
Subgroup 3						
High temperature operation		T_A = 150°C				
Reverse current	4016	dc method V_R = column 11 of table V	I_{R2}		500	µA dc
Subgroup 4						
Small-signal breakdown impedance	4051	I_Z = column 5 of table V; I_{sig} = 10% I_Z rms	Z_Z		Column 6	ohms
Knee impedance	4051	I_{ZK} = 1 mA dc, I_{sig} = 10% I_{ZK} rms	Z_{ZK}		Column 7	ohms
Subgroup 5						
Not applicable						
Subgroup 6		JANS level only				
Surge current (see 4.5.1)	4066	T_C = 25°C; I_{ZSM} = column 10 of table V				
End point electrical measurements		See table IV, steps 1, 3, and 4				
Subgroup 7		JANS level only				
Voltage regulation (see 4.5.2)			$V_{Z(reg)}$		Column 9	V dc
Temperature coefficient of breakdown voltage (see 4.5.4)	4071	I_Z = column 5 of table V; T_1 = 30°C; T_2 = T_1 +100°C	αV_Z		Column 14	/°C

1/ For sampling plan, see MIL-S-19500.
2/ Column references are to table V herein.

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TABLE IIa. Group B inspection for JANS devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Solderability	2026	Dwell time = 10 ±1 s; immersion to cover flat portion of terminal only
Resistance to solvents	1022	
Subgroup 3		
Thermal shock (temperature cycling)	1051	
Hermetic seal	1071	
a. Fine b. Gross		
Electrical measurements		See table IV, steps 1, 3, 4, 5, and 6
Die shear	2017	
Subgroup 4		
Intermittent operating life	1037	I_Z = column 8 of table V, $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. T_C = 30 ±3°C
Electrical measurements		See table IV, steps 1, 3, 4, 5, and 6
Subgroup 5		
Accelerated steady - state operation life	1027	I_Z = column 15 of table V for 96 hours, T_A = 125°C or adjusted, as required, to give an average lot T_J = 225°C
Electrical measurements		See table IV, steps 2, 3, 4, 5, and 6
Subgroup 6		
Thermal resistance	4081	$R_{\theta JC}$ = 12°C/W maximum, T_C = 30 ±3°C. For purposes of this test "junction to case" shall be used in lieu of "junction to lead" and "$R_{\theta JC}$" shall be used in lieu of "$R_{\theta JL}$". The case shall be the reference point for calculation of junction to case thermal resistance ($R_{\theta JC}$). The mounting arrangement shall be with heat sink to case.

1/ For sampling plan, see MIL-S-19500.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Solderability	2026	Dwell time = 10 ±1 s; immersion to cover flat portion of terminal only
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	
Surge current (see 4.5.1)		I_{ZSM} = column 10 of table V
Hermetic seal	1071	
a. Fine leak b. Gross leak		
Electrical measurements		See table IV, steps 1, 3, and 4
Subgroup 3		
Steady-state operation life	1027	I_Z = column 15 of table V T_C = 150 ±5°C
Electrical measurements		See table IV, steps 2, 3, and 4
Subgroup 4		
Decap internal visual (design verification)	2075	
Subgroup 5		
Thermal resistance	4081	$R_{\theta JC}$ = 12°C/W maximum; T_C = 30 ±3°C. For purposes of this test "junction to case" shall be used in lieu of "junction to lead" and "$R_{\theta JC}$" shall be used in lieu of "$R_{\theta JL}$". The case shall be the reference point for calculation of junction to case thermal resistance ($R_{\theta JC}$). The mounting arrangement shall be with heatsink to case.
Subgroup 6		
High temperature (non-operating life)	1032	T_A = 200°C
Electrical measurements		See table IV, steps 2, 3, and 4

1/ For sampling plan, see MIL-S-19500.

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TABLE III. Group C inspection for all quality levels.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Physical dimensions	2066	See figure 1				
Subgroup 2						
Thermal shock (glass strain)	1056					
Terminal strength	2036					
Tension		Test condition A; 20 pounds; t = 15 ±3 s				
Torque (terminal)		Test condition D_1; 10 ounce-inches; t = 15 ±3 s				
Torque (stud)		Test condition D_2; 15 pound-inches; t = 30 ±3 s				
Bending stress		Test condition F; 3 pounds; t = 15 ±3 s, method B.				
Hermetic seal	1071					
a. Fine leak b. Gross leak						
Moisture resistance	1021					
Electrical measurements		See table IV, steps 1, 3, 4, 5, an 6 (JANS) and steps 1, 3, and 4 (JAN, JANTX, and JANTXV)				
Subgroup 3						
Shock	2016					
Vibration, variable frequency	2056					
Constant acceleration	2006					
Electrical measurements		See table IV, steps 1, 3, 4, 5, and 6 (JANS) and steps 1, 3, and 4 (JAN, JANTX and JANTXV)				
Subgroup 4						
Salt atmosphere (corrosion)	1041					

See footnotes at end of table.

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seating plane

5. Measured in the zone .025(0.64 mm) and .050(0.13 mm) from th

TABLE III. Group C inspection for all quality levels) - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 2/						
Barometric pressure reduced (altitude operation)	1001	8 mm Hg				
Subgroup 6						
Steady-state operation life	1026	I_Z = column 15 of table IV T_C = +150 ±5°C				
Electrical measurements		See table IV, steps 2, 3, 4, 5, and 6 (JANS) and steps 2, 3, and 4 (JAN, JANTX, and JANTXV)				
Subgroup 7 3/		JAN, JANTX, and JANTXV levels only				
Temperature coefficient of breakdown voltage (see 4.5.3)	4071	I_Z = column 5 of table V T_1 = 30 ±3°C; T_2 = T_1 +100°C each sublot	αV_Z		Column 14 of table IV	/°C
Voltage regulation (see 4.5.2)		Each sublot	$V_{Z(reg)}$		Column 9 of table V	V dc

1/ For sampling plan, see MIL-S-19500.
2/ LTPD = 15, small lot = 6 devices, C = 0.
3/ LTPD = 10, small lot = 12 devices, C = 0.

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TABLE IV. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits 1/		Unit
		Method	Conditions		Min	Max	
1	Reverse current	4016	DC method; V_R = column 11 of table V	I_{R1}		Column 12	A dc
2	Reverse current	4016	DC method; V_R = column 11 of table V	I_{R3}		Column 13	A dc
3	Breakdown voltage	4022	I_Z = column 5 of table V	V_Z	Column 3	Column 4	V dc
4	Small-signal breakdown impedance	4051	I_Z = column 5 of table V; I_{sig} = 10% I_{Zrms}	Z_Z		Column 6	ohms
5	Knee impedance	4051	I_{ZK} = 1 mA dc I_{sig} = 10% I_{ZKrms}	Z_{ZK}		Column 7	ohms
6	Forward voltage	4011	I_F = 2 A dc	V_F 2/		±50 mV dc change from previously measured value	

1/ Column references are to table V.
2/ Devices which exceed the group A limits for this test shall not be accepted.

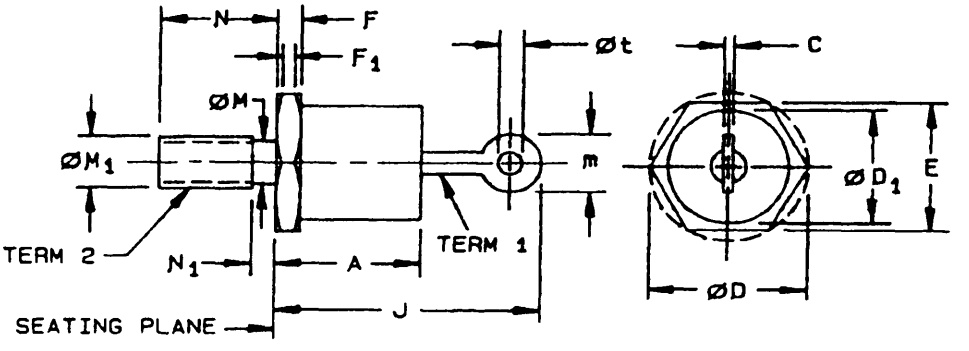
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TABLE V. Test ratings 1/

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15
Voltage group	V_Z Nom	V_Z Min	V_Z Max	I_Z Test current max	Z_Z Impedance max	Z_{ZK} Knee impedance	I_{ZM} 2/ Max dc current T_C = 30 +3°C	$V_{Z(reg)}$ Voltage regulation	I_{ZSM} T_C = 30 ±3°C max	V_R Reverse voltage	I_{R1} Reverse current dc max	I_{R3} Reverse current dc max (after life test)	αV_Z Temperature coefficient max	I_Z Max dc current T_C = 150 +5,-0°C
	volts	volts	volts	mA	ohms	ohms	mA	volts	mA	volts	µA	µA	%/°C	mA
1N3993A	3.9	3.70	4.10	640	2.0	400	2440	0.80	12000	.5	100	200	-.060	610
1N3994A	4.3	4.08	4.52	580	1.5	400	2200	0.72	11000	.5	100	200	-.050	550
1N3995A	4.7	4.45	4.94	530	1.2	550	2000	0.68	10000	1.0	50	100	+.025	500
1N3996A	5.1	4.84	5.36	490	1.1	550	1840	0.64	9000	1.0	10	20	+.030	460
1N3997A	5.6	5.32	5.88	445	1.0	600	1680	0.60	8000	1.0	10	20	+.040	420
1N2970B	6.8	6.46	7.14	370	1.2	500	1520	0.56	7000	2.0	10	20	+.045	380
1N2971B	7.5	7.13	7.87	335	1.3	250	1350	0.45	6750	5.7	100	200	+.057	370
1N2972B	8.2	7.79	8.61	305	1.5	250	1180	0.5	5900	6.2	50	100	+.070	335
1N2973B	9.1	8.65	9.55	275	2.0	250	1100	0.55	5500	6.9	25	50	+.075	305
1N2974B	10	9.50	10.50	250	3.0	250	980	0.6	4900	7.6	25	50	+.081	275
1N2975B	11	10.45	11.55	230	3.0	250	890	0.7	4450	8.4	10	20	+.085	250
1N2976B	12	11.40	12.60	210	3.0	250	820	0.8	4100	9.1	10	20	+.079	230
1N2977B	13	12.35	13.65	190	3.0	250	750	0.85	3750	9.9	10	20	+.080	210
1N2979B	15	14.25	15.75	170	3.0	250	640	1.0	3200	11.4	10	20	+.082	190
1N2980B	16	15.20	16.80	155	4.0	250	605	1.1	3000	12.2	10	20	+.083	170
1N2982B	18	17.10	18.90	140	4.0	250	525	1.25	2600	13.7	10	20	+.085	155
1N2984B	20	19.0	21.0	125	4.0	250	480	1.35	2400	15.2	10	20	+.086	140
1N2985B	22	20.9	23.1	115	5.0	250	435	1.5	2180	16.7	10	20	+.087	125
1N2986B	24	22.8	25.2	105	5.0	250	400	1.65	2000	18.2	10	20	+.088	115
1N2988B	27	25.7	28.3	95	7.0	250	340	1.92	1700	20.6	10	20	+.090	95
1N2989B	30	28.5	31.5	85	8.0	300	320	2.15	1600	22.8	10	20	+.091	85
1N2990B	33	31.4	34.6	75	9.0	300	300	2.30	1500	25.1	10	20	+.092	75
1N2991B	36	34.2	37.8	70	10.0	300	260	2.45	1300	27.4	10	20	+.093	70
1N2992B	39	37.1	40.9	65	11.0	300	240	2.55	1200	29.7	10	20	+.094	65
1N2993B	43	40.9	45.1	60	12.0	400	220	2.75	1100	32.7	10	20	+.095	60
1N2995B	47	44.7	49.3	55	14.0	400	200	3.0	1000	35.8	10	20	+.095	55
1N2997B	51	48.5	53.5	50	15.0	500	185	3.1	925	38.8	10	20	+.096	50
1N2999B	56	53.2	58.8	45	16.0	500	170	3.3	850	42.6	10	20	+.096	45
1N3000B	62	58.9	65.1	40	17	600	150	3.5	750	47.1	10	20	+.097	40
1N3001B	68	64.6	71.4	37	18	600	137	3.8	685	51.7	10	20	+.097	37
1N3002B	75	71.3	78.7	33	22	600	125	4.4	625	56.0	10	20	+.098	33
1N3003B	82	77.9	86.1	30	25	700	115	4.7	575	62.2	10	20	+.098	30
1N3004B	91	86.5	95.5	28	35	800	97	5.1	485	69.2	10	20	+.099	28
1N3005B	100	95.0	105.0	25	40	900	91	6.1	450	76.0	10	20	+.110	25
1N3007B	110	104.5	115.5	23	55	1100	82	6.9	410	83.6	10	20	+.110	23
1N3008B	120	114.0	126.0	20	75	1200	77	7.9	380	91.2	10	20	+.110	20
1N3009B	130	123.5	136.5	19	100	1300	71	8.8	350	98.8	10	20	+.110	19
1N3011B	150	142.5	157.5	17	175	1500	62	11.0	310	114.0	10	20	+.110	17
1N3012B	160	152.0	168.0	16	200	1600	58	11.8	290	121.6	10	20	+.110	16
1N3014B	180	171.0	189.0	14	260	1850	52	13.8	260	136.8	10	20	+.110	14
1N3015B	200	190.0	210.0	12	300	2000	46	14.8	230	152.0	10	20	+.110	12

1/ The test ratings are applicable to all devices.
2/ Derate I_Z linearly to 0.0 mA dc at 175°C, for T_C > 55°C.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.300	.405	7.62	10.29	
C	.012	.065	.30	1.65	
ϕD	----	.505	----	12.83	
ϕD_1	.255	.424	6.48	10.77	
E	.423	.438	10.74	11.13	
F	.075	.175	1.90	4.44	
F_1	.060	.175	1.52	4.44	2
J	.600	.800	15.24	20.32	
ϕM	.163	.189	4.14	4.80	1
ϕM_1	----	----	----	----	5
m	----	.250	----	6.35	3
N	.422	.453	10.72	11.51	
N_1	----	.078	----	1.98	4
ϕt	.060	.095	1.52	2.41	
W	----	----	----	----	

NOTES:
1. Complete threads to extend to within 2-1/2 threads of seating plane.
2. Chamfer on undercut on one or both ends of hexagonal base is optional.
3. Angular orientation of this terminal is undefined.
4. Length of incomplete or undercut threads of ϕM.
5. 10-32 UNF-2A maximum pitch diameter of plated threads shall be basic pitch diameter (.1697 inch (4.310 mm)) reference. (Screw thread standards for Federal Services 1957) Handbook H28 P1.
6. Metric equivalents are given for general information only.

FIGURE 1. Physical dimensions.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Surge current I_{ZSM}. The currents specified in column 10 of table V shall be applied in the reverse direction and shall be superimposed on the current (I_Z = column 5 of table V) a total of five surges at 1-minute intervals. Each individual surge shall be a 1/2 square wave pulse of 1/120-second duration or an equivalent 1/2 sine wave with the same effective (rms) current.

4.5.2 Voltage regulation V_Z(reg). A current at 10 percent of I_{ZM} (column 8) shall be maintained until thermal equilibrium is obtained, and the I_Z shall then be increased to a level of 50 percent of I_{ZM} (column 8) and maintained at this level for a period of time until thermal equilibrium is obtained at which time the voltage change shall not exceed column 9 of table V. During this test, the case temperature (T_C) of the diode shall be equal to 30 ±3°C.

4.5.3 Regulator voltage. The I_Z test current (column 5 of table V) shall be applied until thermal equilibrium is obtained prior to reading the regulator voltage. During this test, the case temperature (T_C) of the diode shall be equal to 30 ±3°C.

4.5.4 Temperature coefficient of regulator voltage (αV_Z). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified case temperatures.

4.5.5 Inspection condition. Unless otherwise specified in MIL-S-19500 or herein, all inspections shall be made at case temperature (T_C) of 30 ±3°C.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Procurement documents should specify the following:

a. Lead material and finish (see 3.2.1).

b. Type designation, polarity type, and product assurance level.

6.3 Part number information. Part numbers 1N3993A, RA through 1N3998A, and RA from MIL-S-19500/272 have been incorporated into MIL-S-19500/124.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 19, 85
DLA - ES
NASA - LRC, MSF

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Preparing activity:
Army - ER

Agent:
DLA - ES

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